$$J_m(U,V) = \sum_{i=1}^{n} \sum_{k=1}^{c} u_{ik}^m \|\mathbf{x}_i - \mathbf{v}_k\|^2, \qquad \sum_{k=1}^{c} u_{ik} = 1$$

$$\Delta x_{jd}^{(t)} = \omega(t)\, \Delta x_{jd}^{(t-1)} + c_1 r_{1jd}(p_{jd} - x_{jd}^{(t-1)}) + c_2 r_{2jd}(g_d - x_{jd}^{(t-1)})$$
$$x_{jd}^{(t)} = x_{jd}^{(t-1)} + \Delta x_{jd}^{(t)}$$

$$\omega(t) = \omega_{\text{đầu}} - (\omega_{\text{đầu}} - \omega_{\text{cuối}})\frac{t-1}{T-1}, \qquad 0{,}9 \rightarrow 0{,}4$$

PSO-S: $\omega(t) = \omega_s - (\omega_s - \omega_e)\left(\frac{t}{T}\right)^2$

PSO-LOG: $\omega(t) = \left[\ln\left(2{,}98 + \frac{t}{T}\right)\right]^{-z}$

$$m_{\text{best},d} = \frac{1}{N}\sum_{j=1}^{N} p_{jd}, \qquad \varphi_{jd} = \rho_{jd}\, p_{jd} + (1 - \rho_{jd})\, g_d, \qquad \rho_{jd} \sim \mathcal{U}(0,1)$$

$$x_{jd} \leftarrow \varphi_{jd} \pm \gamma\, |m_{\text{best},d} - x_{jd}|\, \ln(1/q_{jd}), \qquad q_{jd} \sim \mathcal{U}(0,1], \qquad \gamma = 0{,}9\,\frac{T - t}{T} + 0{,}1$$

$$d_{ik}^2(\boldsymbol{\lambda}) = \sum_{j=1}^{D} \frac{\lambda_j}{\bar{\lambda}} (x_{ij} - v_{kj})^2, \qquad \boldsymbol{\lambda} \in [0{,}05;\ 1]^D$$

$$f(\boldsymbol{\lambda}) = -\,\mathrm{Sil} + P, \qquad P = \frac{\max\,(0,\ 0{,}05 - n_{\min}/n)}{0{,}05}$$

$$\Delta r = \max_k r_k - \min_k r_k, \qquad \mathrm{AUC} = \frac{1}{n_1 n_0} \sum_{y_i=1} \sum_{y_j=0} \left(\mathbb{1}[q_i > q_j] + \tfrac{1}{2} \mathbb{1}[q_i = q_j] \right)$$